

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                               | Marks available |          |          |          |          |          |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) | (i)  |  | (Positive) gravitropism/ geotropism (1)<br>Reject negative gravitropism                                                                                                                                                                                                       |                 | 1        |          | 1        |          | 1        |
|          |     | (ii) |  | Auxin (1)                                                                                                                                                                                                                                                                     | 1               |          |          | 1        |          | 1        |
|          | (b) |      |  | <ul style="list-style-type: none"><li>• {Root/ it} (grows) {straight / horizontal}(1)</li><li>• (Force of) gravity {is equal on/ affects} all sides (of germinating root)/ owtte<br/>direction of gravity is constantly changing/<br/>auxins are spread equally (1)</li></ul> |                 | 1        | 1        | 2        |          | 2        |
|          | (c) |      |  | To show the effect was only because of gravitropism /<br>to test only one variable/<br>eliminate the effect of light/<br>prevent phototropism (1)                                                                                                                             |                 |          | 1        | 1        |          | 1        |
|          |     |      |  |                                                                                                                                                                                                                                                                               | 1               | 2        | 2        | 5        | 0        | 5        |
|          |     |      |  | <b>Total mark for question 4</b>                                                                                                                                                                                                                                              | <b>1</b>        | <b>2</b> | <b>2</b> | <b>5</b> | <b>0</b> | <b>5</b> |